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Vibe Coding

I just see stuff, say stuff, run stuff,
and copy paste stuff, and it mostly works

A. Karpathy (OpenAI)

Underlying problem

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Servizio | Nuove tecnologie

InvestCloud di Marghera licenzia tutti i 37 dipendenti, sostituiti dall'Ai

Il nuovo modello organizzativo del gruppo statunitense basato su sistemi integrati con l'intelligenza artificiale non prevederebbe il mantenimento di strutture locali autonome

di Barbara Ganz

11 marzo 2026

TECH • AI

Employment for computer programmers in the U.S. has plummeted to its lowest level since 1980—years before the internet existed

By Sasha Rogelberg

Reporter

March 17, 2025, 2:00 PM ET

Why 90% of Programmers Will Be Unemployed by 2030 (The AI Takeover) | Dev Tech Insights

ABDUL REHMAN KHAN / APRIL 12, 2025 / TECH NEWS

Introduction

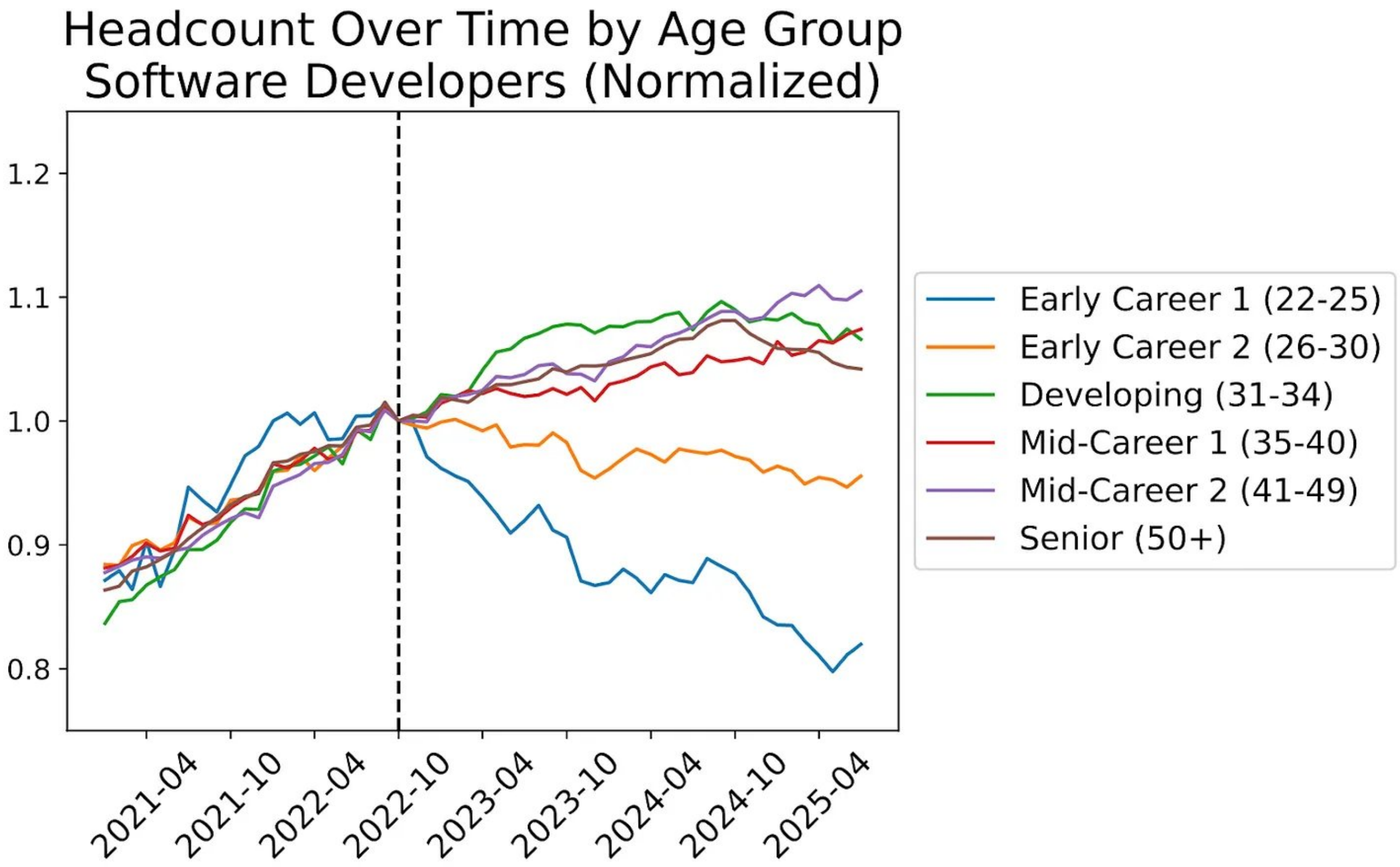
The programming profession is facing an existential crisis.

In 2022, GitHub Copilot shocked the world by automating simple coding tasks.

In 2024, AI agents like Devin demonstrated the ability to build full applications from scratch.

By 2030, advanced AI will make **90% of traditional programming jobs obsolete**.

Underlying problem



Why?

- AI is taking over traditional programming
- Developers must learn to ride the AI wave
  - **Or accept being replaced!**
- From development to “code review”
  - Vibe Coding and beyond

Question: is it still necessary to know how to program?

What is Vibe Coding?

- Credits: Andrej Karpathy (Feb. 2025)
- You simply describe what do you want as input for an AI engine
  - You do not have to write code
  - You do not understand what is under the hood
  - Anyway, if it works it can be ok!

Most used tools

- ChatGPT / Claude
- GitHub Copilot
- Cursor / Windsurf
- Replit Agent

Practical Vibe Coding Flow

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1

DESCRIBE

“Write a C program that sort an array of integers”

2

AI OUTPUTS CODE

AI produces full code with #include, main(), etc.

3

EXECUTE

You have to compile and execute. If it fails, paste error or describe malfunctioning back to AI console

The cycle repeats: the user never needs to understand the code, only verify the result.

Welcome to the dark side!

Speed

In seconds you get code that compiles and runs. No more hours hunting for a missing semicolon.

No syntax needed

You speak your mother's tongue, AI translates to C. No need to remember printf, scanf, pointers...

"Easy" debugging

Copy the compiler error, paste it in the chat, AI fixes it. Feels like magic.

Instant result

The program works! You submit it. Exam passed... or maybe not?

The problem: what you DON'T learn

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With Vibe Coding you lose...

- How memory actually works
- Algorithmic reasoning
- The ability to debug on your own
- Understanding pointers and arrays
- Computational thinking

With proper study you gain...

- Mastery of C's memory model
- Structured problem solving
- Autonomy in finding and fixing errors
- Ability to read and write real code
- Solid foundations for any future language

Practical example: sorting an array

The "vibe coder" student says:

"Write a C program that sorts an array of 10 integers"

AI generates → it works! But...

```
void sort(int a[], int n) {
  for(int i=0; i<n-1; i++)
    for(int j=0; j<n-i-1; j++)
      if(a[j] > a[j+1]) {
        int t = a[j];
        a[j] = a[j+1];
        a[j+1] = t;
      }
}
```

The prepared student knows:

What algorithm is this?	Bubble sort — $O(n^2)$
Why j < n-i-1?	Last elements are already sorted
How does the swap work?	Temp variable (memory model)
Better alternatives?	Quicksort, Mergesort...

The vibe coder has code that works. The prepared student has code they understand.

The real risks of Vibe Coding

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Security

Generated code often has vulnerabilities: buffer overflows, missing input validation, memory leaks. The AI won't warn you.

Hidden bugs

The program seems to work but has undefined behavior. Without understanding C, you don't even know where to look.

Skill atrophy

If you don't practice, you don't learn. Your brain needs active exercise, not passive observation of generated code.

At the exam: alone

At the exam you won't have ChatGPT. You'll need to write C from scratch, find bugs, reason about algorithms. No AI help.

The Evolution:
from Vibe Coding to Agentic Engineering

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February 2025

Vibe Coding

"Forget that the code exists"

Good for: prototypes, weekend projects, quick experiments

February 2026

Agentic Engineering

"Oversight and quality without compromise"

Requires: expertise, oversight, code comprehension

The message is clear: **even Karpathy** acknowledges that pure vibe coding isn't enough. To use AI well you need **solid technical expertise.**

Unsing AI in the proper way

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1. First: learn the fundamentals

C syntax, types, memory, pointers, basic algorithms. Write everything by hand. Make mistakes. Fix them. Repeat.

2. Then: use AI as a tutor

Ask for explanations: "Why does this code segfault?", "How does malloc work?". AI explains, you learn.

3. Finally: use AI as an accelerator

Once you master C, AI helps you be faster. But you can read, verify, and fix its output.

AI is incredibly powerful, but only if you know how to guide it.

Final remarks

- Don't use AI to avoid learning
- Use it to learn better

Take Away

Vibe coding is the junk food of programming: **instant gratification, but no nourishment.**

- Invest time in fundamentals now:
  - it will pay off for your entire career.

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Vibe Coding

No AI was harmed in the making of this presentation

But it is pure AI stuff…